RESEARCH ARTICLE

Enhanced Multistep Prediction of Multivariate Market Indices Using Weighted Optical Reservoir Computing

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Abstract
We propose and experimentally demonstrate an innovative weighted optical reservoir computing system for market index prediction. By integrating fundamental market data with macroeconomic indicators and technical metrics, we capture the broader dynamics of the stock market. Our system shows significantly higher performance than the state-of-the-art methods such as linear regression, decision trees, and neural network architectures including long short-term memory. It effectively captures the market's high volatility and nonlinear behaviors under limited data conditions, demonstrating strong potential for real-time, parallel, multi-dimensional data processing, and prediction.

Keywords Optical reservoir computing · Parallel data processing · Multivariate forecasting · Multistep prediction

1 Introduction

Over the past few decades, machine learning (ML) has emerged as a transformative tool in financial market prediction, particularly for forecasting market indices [1]. The core challenge in this field lies in effectively integrating diverse information sources and building models capable of capturing the inherent complexity of financial markets. A wide range of ML techniques-including regression, classification, deep learning, and ensemble approaches-have been applied to forecast stock and market trends [2–9]. Among them, long-short-term memory (LSTM) networks, a specialized class of recurrent neural networks (RNNs), have gained prominence for their ability to capture temporal dependencies and nonlinear patterns in sequential data [10–16]. Leveraging historical prices, technical indicators [17], and macroeconomic variables, these ML models can generate nuanced forecasts and provide insights into market behavior.

Despite their promise, existing ML methods for market prediction face several limitations. Most require large volumes of dense historical data for training, which can be resource-intensive and not always practical. Moreover, many models struggle to capture sudden market shifts or peaks, often leading to unreliable predictions under volatile conditions. These limitations hinder their effectiveness for real-time forecasting and financial decision-making.

RNNs have long served as foundational architectures for modeling temporal dynamics, but their training is often challenged by issues such as vanishing and exploding gradients. To address these issues, alternative frameworks have been developed. One such approach is Backpropagation-Decorrelation (BPDC), an efficient online learning algorithm introduced by Steil [18], which reduces computational complexity to $\mathcal{O}(N)$ while enhancing training stability. Another influential direction is Reservoir Computing (RC), which simplifies learning by keeping the recurrent component-known as the reservoir-fixed and training only the output layer [19]. A prominent RC model is the Echo State Network (ESN), proposed by Jaeger [20], which employs a randomly initialized reservoir that

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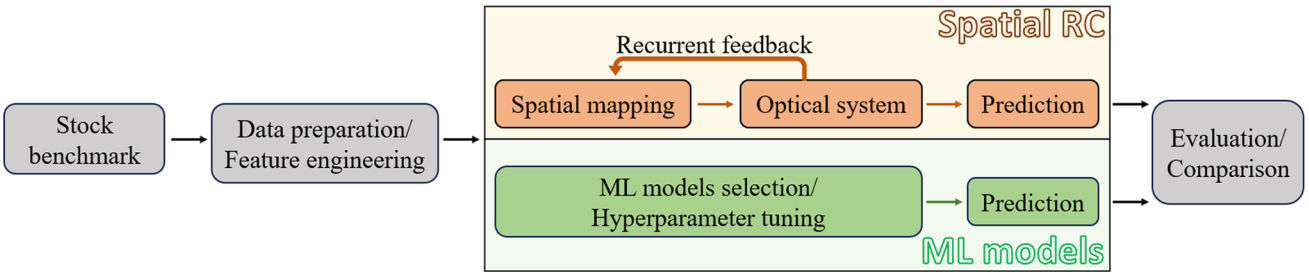


Fig. 1 Experimental setup for stock index prediction comparing spatial reservoir computing and popular machine learning models

satisfies the echo state property to ensure stability. These architectures offer a compelling balance of expressiveness, computational efficiency, and ease of training for time-series modeling.

Recent studies have demonstrated that RC architectures can perform competitively on time-series tasks, even with limited training data [21–23]. This makes them particularly attractive for financial applications where low-latency, real-time, and data-efficient prediction is critical. An RC system consists of a dynamic reservoir that projects input into a high-dimensional space, and a readout layer that extracts patterns from the resulting reservoir states [24]. As only the readout is trained-typically with simple linear regression-RC allows for fast training and efficient computation [25]. Furthermore, the fixed nature of the reservoir makes RC highly suitable for hardware implementation. In particular, optical reservoir computing has attracted attention for its ability to leverage photonic hardware for ultra-fast and parallel computation [23, 26–28]. Optical RC systems can be categorized into free-space optical RC, fiber-based temporal RC, and integrated photonic RC, with free-space configurations offering the highest flexibility and parallelism.

In this work, we propose and experimentally demonstrate an innovative free-space optical RC system for forecasting daily stock index prices. Using a spatial light modulator (SLM), temporal benchmark data is projected into high-dimensional optical signals across both spatial and temporal domains. This architecture enables high-quality prediction while supporting flexible encoding of multi-dimensional input features. Thanks to the large number of modulator pixels and a novel feature-mapping method, the system is capable of processing multiple training samples in parallel-regardless of input feature size-without incurring additional time costs for signal interconnection or nonlinearity, as seen in traditional temporal RC systems [22].

By translating temporal dependencies in time-series data into spatial coherence within a free-space optical platform, our system enables multi-step prediction with enhanced parallelism and speed. We evaluate performance on seven widely-used stock indices, each with 500 historical data points and seven macroeconomic and technical indicators. Feature correlation is incorporated as a weighting mechanism to further enhance prediction quality. Our comparative analysis shows that the proposed optical RC system consistently outperforms popular ML models for both one-step and multi-step prediction tasks. Unlike prior approaches that depend on large datasets, our system achieves real-time, low-latency, and accurate forecasting with limited historical data. Notably, the integration of feature-weighted mapping leads to immediate capture of signal peaks and improved robustness. These advantages position our feature-optimized optical RC system as a promising solution for edge-based confidential forecasting and efficient financial modeling.

2 Methods

Experiment Diagram

Figure 1 illustrates the experimental procedures described in this paper. Initially, we download historical trading datasets from stock benchmarks and perform basic data and feature engineering. Subsequently, we apply optical

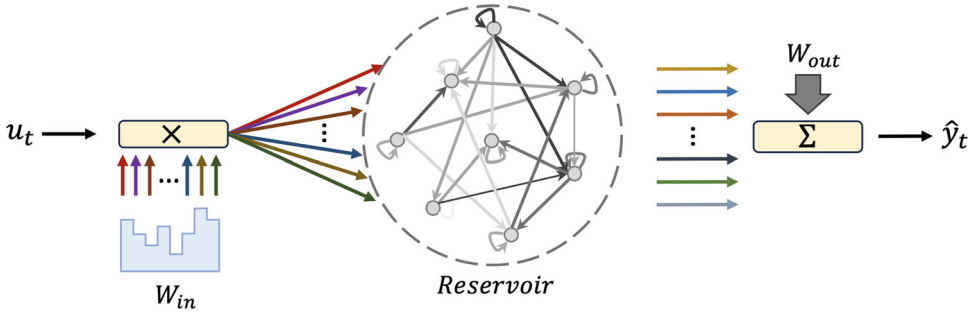


Fig. 2 The schematic diagram of the conventional RC

RC and various popular ML techniques to predict future stock index prices, respectively. We compare different machine learning models, utilizing cross-validation and hyperparameter tuning to select the ML model with the lowest error for each benchmark. Finally, we evaluate our models and compare the predictions from optical RC with the best-performing ML model for each benchmark.

Optical RC

Our spatial optical RC follows the basic working principle of conventional RC as visualized in Fig. 2, which can be described by

$$x(t) = f[\alpha x(t-1)W_{res} + \beta u(t)W_{in}], \tag{1}$$

where $x(t)$ represents the reservoir state at time t. f is the nonlinear function. α and β are the feedback gain and input gain, which will be optimized for the best performance. W_{res} is the random connection of the reservoir nodes. W_{in} is the input weight to map the input to the system. In a conventional RC framework, the input vector x has a dimension of $L \times 1$, where L denotes the length of the time series. At each time step t, there is a single data point. The input weight matrix W_{in} is generated randomly with dimensions $N \times L$, where N is the number of reservoir nodes. This setup enables the projection of the input data into a higher-dimensional space. Due to the updating rule of the reservoir states in Eq. 1, the current state $x(t)$ inherently contains information from the preceding state $x(t-1)$. This endows reservoir computers with the crucial property of short-term memory, a feature for effective time-series data prediction. In the training stage, the input is injected sequentially into the reservoir, and the resulting reservoir states x for all time instances within the range [1, T] are recorded. Subsequently, the system's output weights (W_{out}) can be determined by minimizing the error between the predicted results ($\hat{y} = W_{out}\boldsymbol{x}$).

In our optical setup, a continuous-wave 775 nm laser diode serves as the light source, as illustrated in Fig. 3. The beam waist is adjusted to 1 mm after collimation. To match the effective area on the spatial light modulator (Santec SLM-210, which is 15.4 × 9.6 mm), a 10× beam expander is used to enlarge the beam size. A power meter is implemented to capture the transmitted light passing through the polarizing beam splitter (PBS) for real-time monitoring of system stability during training and testing. While the laser power is not directly incorporated into the computational process, it is utilized as a diagnostic measure to mitigate the risk of erroneous results arising from abrupt fluctuations in the laser source. Before the beam splitter (BS), we employ a half-wave plate to rotate the light's polarization, thereby maximizing the modulation effect on the SLM. Following modulation, the light is focused, and a camera (model: Basler a2A 1920-160umPRO with a 1920 × 1200 resolution) is positioned in the time-Fourier domain [29] to align the sensor area with the beam size for subsequent information processing.

The inset in Fig. 3 illustrates the data flow diagram in our spatial RC. One key here is the combination of input values and the feedback state for spatial light modulation. Here, we use a similar mapping method as the "linear-combination mapping" in our previous paper [21]. The difference is with how the overall mapping area is split into several input portals. Instead of equally dividing the SLM into n sections for n input values, we repeat and stack the inputs until all the modulation areas on the SLM are filled. In the case where $n = 4$, as shown in

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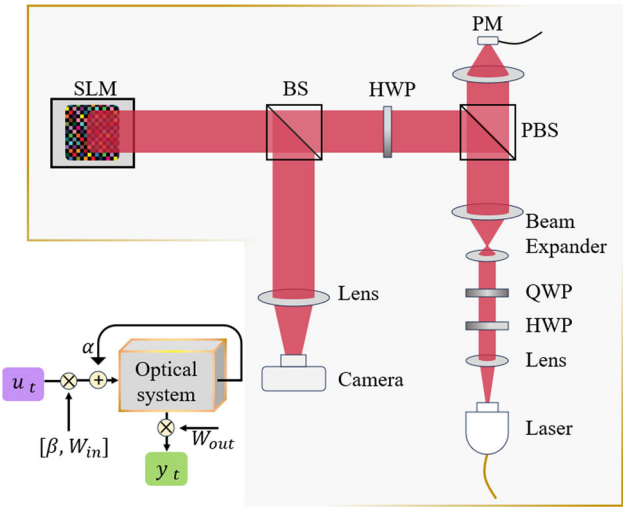


Fig. 3 The configuration of the spatial reservoir computer. SLM: spatial light modulator, BS: beam splitter, HWP: half-wave plate, QWP: quarter-wave plate, PBS: polarizing beam splitter, PM: power meter

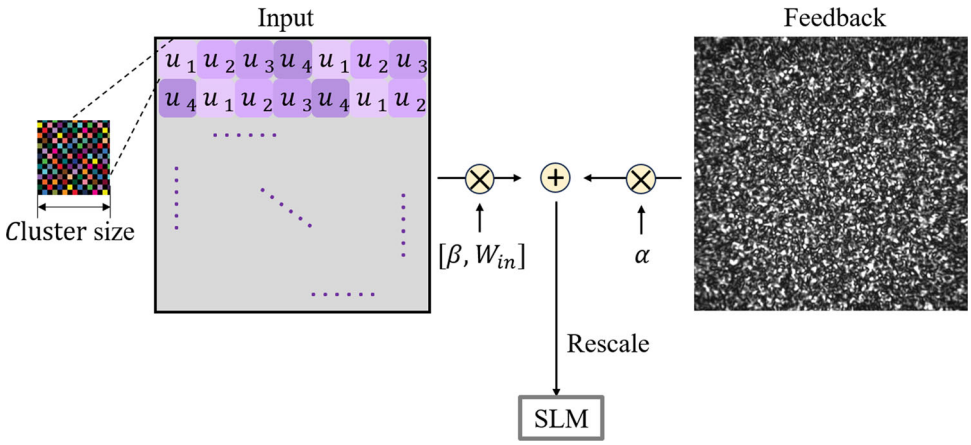


Fig. 4 The mapping method to combine multi-feature inputs and the camera state for spatial light modulation

Fig. 4, the 4 input values are repeated and stacked sequentially, each with a height and length of 20 SLM pixels. The mapped input is multiplied by an input scale (β) and a random input weight(W_{in}), while the captured camera state is rescaled with a feedback gain (α). The combination is then linearly rescaled again to match with the $0-2\pi$ phase modulation on SLM. The nonlinearity of the system is achieved through the camera intensity reading and the intensity saturation rate. Similar to Eq.(1), our spatial RC can be simplified as

$$x(t) = f\{[\alpha x(t-1) + \beta W_{in} \odot W_{cor} \odot u(t)]W_{res}\}. \tag{2}$$

Here W_{cor} is a designed weight for the features of each stock index benchmark. $x(t)$ is the reservoir state at time t from the camera, which is rescaled to match the gray level of the SLM. $\odot$ is a point wise multiplication operator. $f\{\}$ and W_{res} are the accumulated nonlinear effect and reservoir weight, respectively, through the whole system.

Thanks to the large number of pixels on SLM (400×400), it can easily function as a multi-input portal, allowing a large amount of information to be encoded into the system at once. In this way, the temporal correlation among the time-series data is translated into spatial coherence of the optical field, whose dynamics implement a complex-number kernel function. Therefore, besides evaluating the 1-step prediction, where the result contains only the next step value, we also assess the system's performance for multi-step ahead prediction using more inputs. In this case, $u(t)$ is the vector containing all the feature information for different time steps. Specifically, we present examples of multi-step prediction with four-step and ten-step ahead predictions, labeled as 4-step and 10-step, respectively. Taking 4-step prediction as an example, the input vector $u(t)$ has 32 values, corresponding to one closing price and 7 features per time step. Accordingly, the prediction horizon is 4 steps containing 4 days of closing prices beyond

Table 1 Stock indices statistics summary

Stock index	Start date	End date	Trading days	Max price	Min price	Mean price	Std Dev price
^HSI	2021-03-22	2023-09-29	600	29468.00	14687.02	22068.88	3580.74
^IXIC	2021-05-07	2023-09-29	600	16057.44	10213.29	13074.77	1547.05
^N225	2021-03-16	2023-09-29	600	33753.33	24717.53	28589.14	1910.66
^NYA	2021-05-07	2023-09-29	600	17353.76	13472.18	15848.07	836.49
MME=F	2021-05-07	2023-09-29	600	1391.90	846.40	1091.06	144.08
UKX.L	2021-04-20	2023-09-29	600	8012.53	6826.15	7379.49	270.15
VTHR	2021-05-07	2023-09-29	600	216.93	160.90	191.04	13.50

the most recent input. This encoding method exploits the spatial correlation of the free space optics that occurs naturally in our setup. To explore the impact of input size on multi-step forecasting, we conducted supplementary experiments on the Mackey-Glass benchmark using fixed-length input vectors. As this analysis is peripheral to the main focus of the paper, we briefly summarize the findings here for readers who may find them informative: With the same size of input, the performance degraded notably at longer prediction horizons (e.g., NMSE increased from 0.0009 for 1-step prediction to 0.016 in the 4-step case), likely due to limited temporal context.

Multiple ML and LSTM Models

This paper uses various ML regression methods and neural networks to predict each benchmark [30, 31]. These include Lasso, ElasticNet, Random Forest Regressor, Linear Regression, Ridge, SGD Regressor, K-Nearest Neighbors Regressor, Decision Tree Regressor, Bayesian Ridge, and LSTM [32–34]. Specifically, the LSTM architecture includes an internal memory that acts like a local variable, storing information as it processes a sequence of inputs. It uniquely handles outputs by recycling the production from one time-step as the input for the subsequent time-step [35]. This recycling process enables the LSTM to make informed predictions by considering both the current input and the output from the immediate preceding timestep. While LSTM is well-suited for time series prediction due to its ability to capture temporal dependencies, it generally requires large datasets for effective training. To ensure a fair comparison, we also evaluate several traditional machine learning regression models. We employ cross-validation with time series splitting to rigorously evaluate and compare each model's performance while preserving the temporal structure of the data. After that, we apply grid search for hyperparameter tuning to optimize the model's configuration, ensuring we identify the most effective parameters for the best-performing model. The best-performing model will be selected for comparison with the optical RC system.

For multi-step prediction, the data preparation for the time-series sequence with features follows the same approach as in optical RC in the previous subsection. The regression methods remain consistent with the one-step prediction, while the model dimensions in the LSTM are adjusted to align with the prediction horizon.

3 Datasets and Feature Engineering

Market Indices

Table 1 presents statistical data for seven major stock indices over periods ranging from early 2021 to late 2023, sourced from Yahoo Finance. Each index's performance is summarized by the number of trading days, maximum and minimum prices, mean price, and standard deviation of prices. Based on the statistical data, the closing prices of all the stock indices are inherently noisy, non-linear, and chaotic. Indices such as ^HSI and ^N225 exhibited higher volatility, while MME=F and VTHR demonstrated greater price stability. This table provides a comparative overview of the behavior of global equity indices during the examined period.

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Table 2 Features used for stock prediction in this paper

Fundamental data	**Source**	**Abbreviation**
Closing price	Yahoo Finance	Close
Closing price(t-1)	Yahoo Finance	Close(t-1)
Macroeconomic data	**Source**	**Abbreviation**
Cboe volatility index	Yahoo Finance	VIX
Effective Federal Funds Rate	FRED	EFFR
Consumer sentiment index	FRED	UMSCENT
US dollar index	Yahoo Finance	DXYNYB
Technical analysis data	**Source**	**Abbreviation**
Moving average convergence divergence	Calculated	MACD
Average true range	Calculated	ATR
Relative strength index	Calculated	RSI

In dynamic and highly non-stationary Financial environments especially like the stock market, concept drift is a persistent issue: the underlying data-generating processes evolve over time due to changing market regimes, investor behavior, and macroeconomic factors. As a result, models trained on large historical datasets often suffer from degraded predictive performance when applied to more recent data, since the patterns they learn may no longer be relevant [36]. The decision to train our models on relatively small datasets-specifically, 500 daily observations-was made with careful consideration of the unique challenges posed by financial time series. By focusing on shorter, more recent time windows, we aim to mitigate the impact of outdated information and ensure that the model reflects the current market context. The importance of handling concept drift in streaming or time-evolving environments has been well documented in the literature [37], and is especially critical in financial applications where drift can occur both abruptly and incrementally.

Features and Correlation Weights

Due to the complexity of the benchmarks, a prediction model based solely on the closing price cannot gather all the necessary information for accurate predictions. Consequently, feature selection is a crucial part of the prediction enhancement. However, it is challenging to extract features from financial data. On the one hand, limiting the features can hinder the predictive model's performance. On the other hand, including all available features from the financial market can lead to a dimensionally complex and difficult-to-interpret model. Additionally, collinearity among multiple variables can negatively impact the model's performance. Some studies select the features rely solely on technical indicators [38, 39], while others incorporate historical data and macroeconomic indicators [40–44].

Table 2 outlines the set of input features used for stock prediction in this study, categorized into fundamental, macroeconomic, and technical indicators. Fundamental data includes the current and lagged closing prices obtained from Yahoo Finance. Macroeconomic indicators incorporate market sentiment and economic conditions, such as the Cboe Volatility Index (VIX), Effective Federal Funds Rate (EFFR), Consumer Sentiment Index (UMSCENT), and the U.S. Dollar Index (DXYNYB), sourced from Yahoo Finance and FRED. Technical analysis features, including the Moving Average Convergence Divergence (MACD), Average True Range (ATR), and Relative Strength Index (RSI), are calculated directly based on historical price data. These diverse features are designed to capture both short-term dynamics and broader market conditions to enhance forecasting accuracy. The features are selected by calculating the Pearson correlation coefficients, which reflect the strength and direction of the linear relationship between the features and the closing price of each stock index.

All the input features discussed above contribute to stock price prediction. This relationship is illustrated by a correlation matrix, as shown in Fig. 5. The numerical values in the heatmap represent the pairwise correlations between the variables on the horizontal and vertical axes. As expected, the diagonal values of the matrix are all

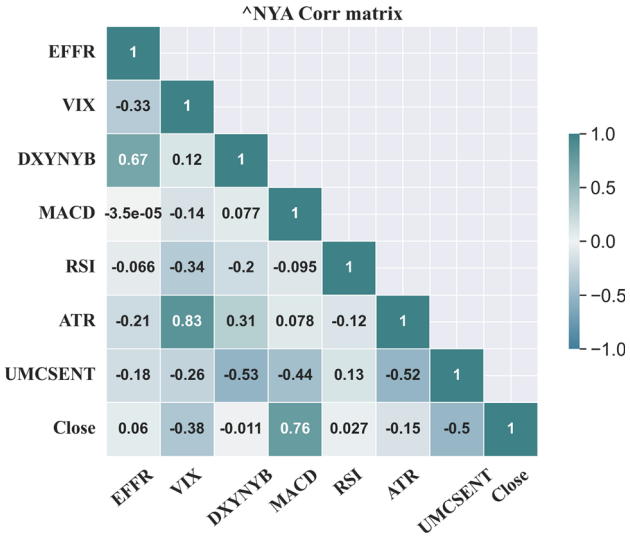


Fig. 5 The correlation weights between the features and the closing price for stock index ^NYA

1, indicating perfect correlation of each variable with itself. Therefore, these diagonal values are not informative for our analysis. Instead, the off-diagonal entries are used in the feature selection process. The intensity of the color in Fig. 5 reflects the magnitude of these correlations, indicating the strength of the relationship between corresponding variables.

In particular, for the subsequent RC prediction, we incorporate the bottom row of the correlation weight (CW) matrix as an additional set of input weights, denoted as W_{cor}, as shown in Eq. (2). These weights are integrated into the input data during the stage where random input weights are applied, serving as scaling factors that modulate the influence of each input indicator in the spatial RC system. This mechanism emphasizes features with stronger correlations (i.e., higher weights) while downweighting less informative ones. By doing so, the input information is selectively emphasized or de-emphasized according to its assigned weight, enabling the system to integrate data in a more discriminative manner, focus more effectively on relevant inputs and ultimately enhancing its predictive effectiveness. All inputs are standardized using z-score normalization prior to training to ensure consistency in scale. Further analysis and validation are presented in the Results and Discussion section.

4 Evaluation Metrics

Table 3 provides a comprehensive overview of the evaluation metrics used to assess the performance of the optical RC and ML models in this study. The selected metrics span several categories to ensure a holistic evaluation. Error-based metrics such as MAE, MSE, RMSE, NRMSE, and MAPE quantify the absolute and relative deviations between predicted and actual values. The coefficient of determination (R^2) captures the proportion of variance explained by the model, while Directional Accuracy(%) measures the ability to predict the correct movement direction.

To evaluate the performance of the proposed approach from a trading perspective, financial performance metrics—including Cumulative Return, Annualized Return, and Maximum Drawdown—are incorporated. Risk-adjusted performance is evaluated using the Sharpe Ratio, Sortino Ratio, and Calmar Ratio, which account for volatility and downside risk. Additional measures such as the Information Ratio, Win Rate, Profit Factor, and Trade Expectancy offer insight into profitability, consistency, and expected outcomes of individual trades. Together, these metrics provide a multidimensional framework for comparing model accuracy, robustness, and practical relevance in financial prediction tasks.

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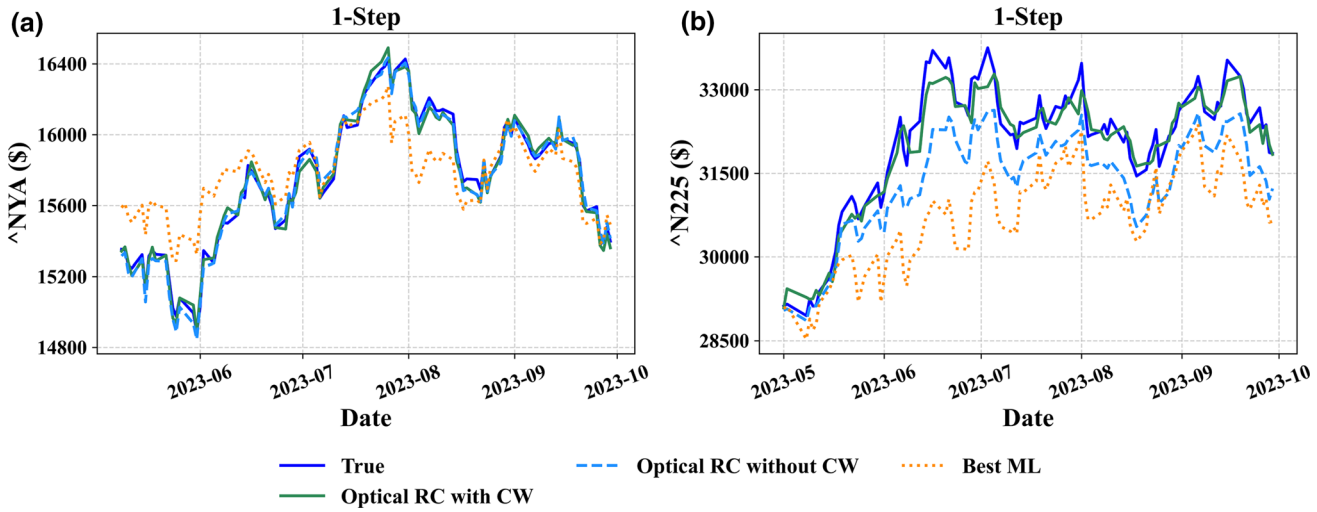


Fig. 6 1-step prediction for **a** ^NYA and **b** ^N225 indices using the optical RC system with correlation weights, without correlation weights and the best-performing ML model

5 Results and Discussion

In our experiments, we used 500 days of data for training and 100 days for testing. Among the ML baseline models, Lasso performed best for 1-step prediction, while LSTM outperformed the other models for 4-step and 10-step predictions. This is because Lasso's L1 regularization helps prevent overfitting under limited data conditions. In contrast, LSTM benefits from its neural network architecture, which is well suited to capturing temporal dependencies-particularly after data resampling during the 4-step and 10-step modeling. In the results that follow, we compare only the best-performing ML models, labeled as Best ML, with the weighted optical RC system, labeled as optical RC.

Figure 6 presents a subset of the experimental results, highlighting the significant performance improvement achieved by incorporating correlation weights into the optical RC system. Figures present 1-step prediction results for the stock indices ^NYA and ^N225, respectively. For ^NYA, the normalized root mean square errors (NRMSEs, the formula is listed in Table 3) of the RC model with and without W_{cor} are 0.104 and 0.108, respectively, while the best ML model yields an NRMSE of 0.558-over five times higher than the RC result. Although the performance difference between the optical RC systems with and without correlation weights is relatively small for ^NYA, the inclusion of W_{cor} still offers measurable improvement. In contrast, the effect of W_{cor} is much more significant for ^N225, as shown in Fig. 6(b). The RC model without correlation weights yields an NRMSE of 0.688, while the inclusion of W_{cor} reduces the error to 0.236-a substantial enhancement. This clearly demonstrates that the feature correlation weights substantially boost the predictive power of the optical RC system, especially in more complex market conditions. Furthermore, the optical RC system consistently outperforms the best ML model across both benchmarks. For ^N225, the top ML model results in an NRMSE of 1.309, significantly worse than both RC configurations. As visually evident in Fig. 6, the ML model predictions lag behind true price movements and diverge from actual values, whereas the optical RC predictions more accurately capture market volatility and nonlinear trends.

This indicates that the inclusion of correlation weights significantly enhances the prediction accuracy of the optical RC system by enabling it to better capture inter-feature relationships. The advantage of the weighted optical RC is especially pronounced in Fig. 6b, where the optical RC with CW not only follows the trend more accurately but also achieves better amplitude matching in the rising and falling phases of the ^N225 index. In contrast, the best ML model exhibits noticeable lag and under-performance, particularly during periods of sharp market movement.

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Table 3 Descriptions and formulas for the evaluation metrics used in model performance analysis

Metric	Description	Formula
MAE	Mean Absolute Error	$\frac{1}{n}\sum_{i=1}^{n}\left\|y_i-\hat{y}_i\right\|$
MSE	Mean Squared Error	$\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2$
RMSE	Root Mean Squared Error	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}$
NRMSE	Normalized Root Mean Squared Error	$\frac{\sqrt{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}}{\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}}$
MAPE	Mean Absolute Percentage Error	$\frac{100}{n}\sum_{i=1}^{n}\left\|\frac{y_i-\hat{y}_i}{y_i}\right\|$
R^2	Proportion of variance in the dependent variable predictable from the independent variables	$1-\frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}$
Directional Accuracy %	Percentage of correct directional predictions	$\frac{1}{n}\sum_{i=1}^{n}\mathbf{1}\left[\text{sign}(\hat{y}_i)=\text{sign}(y_i)\right]$
Cumulative Return	Total return over time from a strategy	$\prod_{t=1}^{n}(1+r_t)-1$
Annualized Return	Average compounded return per year	$(1+R)^{\frac{252}{n}}-1$
Maximum Drawdown	Largest peak-to-trough decline in portfolio value	$\max_t\left(\frac{\text{Peak}_t-\text{Val}_t}{\text{Peak}_t}\right)$
Calmar Ratio	Annualized return divided by maximum drawdown	$\frac{\text{Annual Return}}{\text{Max Drawdown}}$
Sharpe Ratio	Risk-adjusted return	$\frac{\mathbb{E}[R-R_f]}{\sigma_R}\cdot\sqrt{252}$
Sortino Ratio	Risk-adjusted return but penalizes only downside risk	$\frac{\mathbb{E}[R-R_f]}{\sigma_{\text{down}}}\cdot\sqrt{252}$
Information Ratio	Return relative to a benchmark divided by tracking error	$\frac{\mathbb{E}[R-R_b]}{\sigma_{R-R_b}}\cdot\sqrt{252}$
Win Rate	Percentage of trades that are profitable	$\frac{\text{Profitable Trades}}{\text{Total Trades}}$
Profit Factor	Total gains divided by total losses	$\frac{\sum\text{Profits}}{\left\|\sum\text{Losses}\right\|}$
Trade Expectancy	Average expected return per trade	$(\text{Win Rate}\cdot\text{Avg Gain}-(1-\text{Win Rate})\cdot\text{Avg Loss})$

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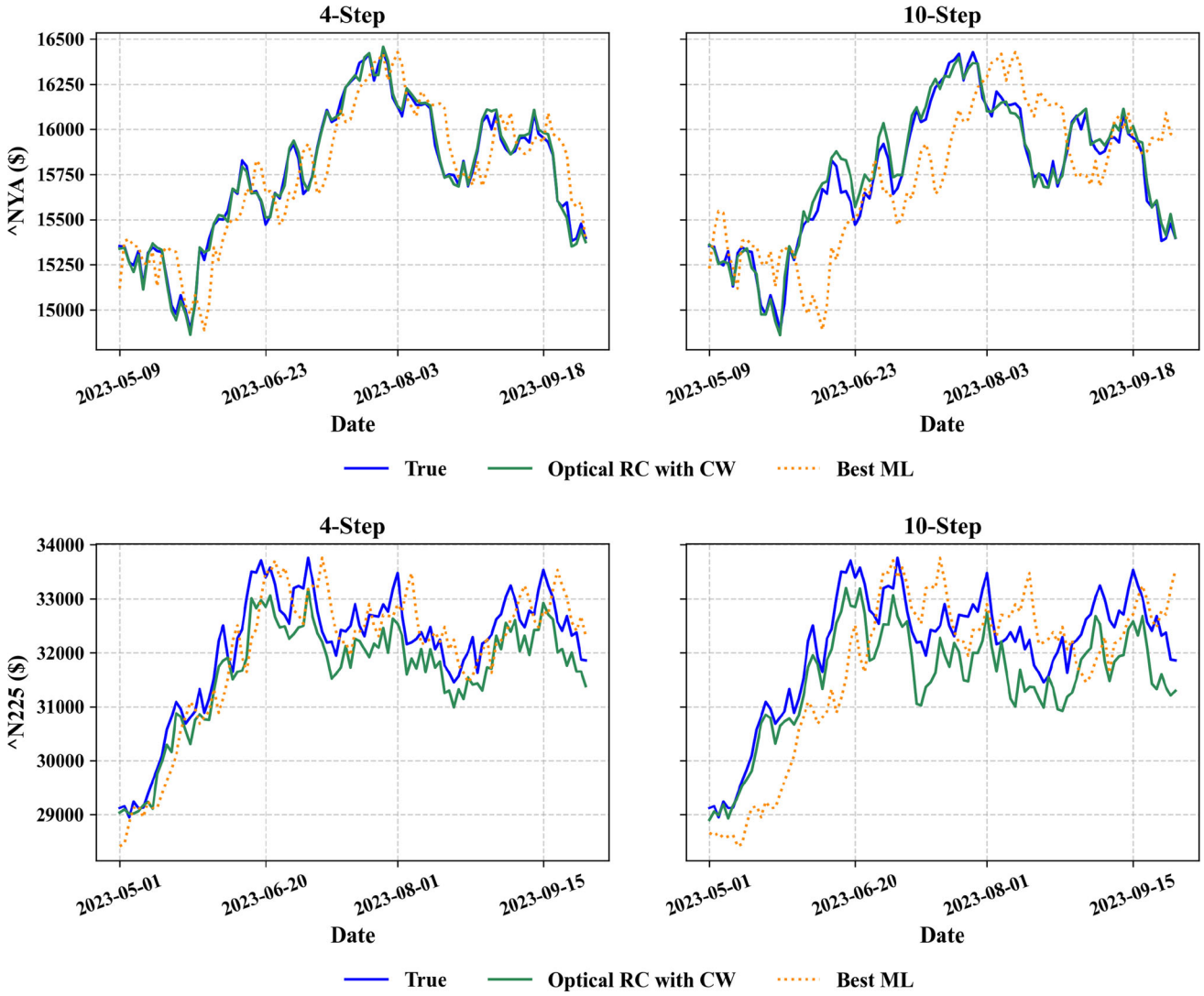


Fig. 7 Multi-step price prediction curves for stock indices ^NYA and ^N225

The figures visually confirm the effectiveness of incorporating correlation weights into the optical RC framework, setting the stage for the weighted optical RC to be the primary model used in all subsequent experiments.

To gain insight into the prediction performances between the weighted optical RC and ML models for multistep predictions, we plot the prediction results. Figure 7 compares the true closing prices with predictions from the optical RC and the best-performing ML model, across 4-step, and 10-step horizons. Similar to the 1-step case, here we take ^NYA and ^N225 as demonstration as well. The figures in the first row are for ^NYA and those in the second row are for ^N225. The figures illustrate that for 4-step predictions, the optical RC continues to preserve the overall trend with stable performance, while ML shows increasing deviation. ML has delayed reaction to all movements, while optical RC maintains closer alignment with the actual trajectory, showing better temporal generalization and adaptability. By the 10-step horizon, the performance gap becomes substantial. Optical RC retains the core trend structure with slightly dampened amplitude-an acceptable trade-off in long-horizon forecasting-whereas ML becomes highly erratic, exaggerating volatility.

To have an all-round analysis on the prediction performance, we calculate the performance based on the statistical metrics as shown in Fig. 3 for both the optical RC model and the best-performing ML model across 1-step, 4-step, and 10-step forecasting horizons. Taking ^NYA index as an example, Table 4 lists all the comparison results. At

Table 4 NYA statistical analysis between the weighted optical RC and the best ML model across multiple prediction steps

	1-step Optical RC	1-step ML	4-step Optical RC	4-step ML	10-step Optical RC	10-step ML
MAE	31.56	165.26	22.76	150.58	47.03	257.42
MSE	1480.29	42944.01	931.30	33892.23	3769.91	97051.14
RMSE	38.47	207.23	30.52	184.10	61.40	311.53
NRMSE	0.10	0.56	0.08	0.50	0.17	0.84
MAPE	0.20	1.06	0.14	0.96	0.30	1.63
R^2	0.99	0.69	0.99	0.75	0.97	0.30
Directional Accuracy %	86.87	94.95	93.94	59.60	79.80	54.55
Cumulative Return	0.02	0.02	0.07	−0.01	0.07	0.07
Annualized Return	0.05	0.07	0.17	−0.01	0.18	0.18
Maximum Drawdown	0.04	0.04	0.03	0.04	0.04	0.04
Calmar Ratio	1.20	1.90	5.29	−0.27	4.51	4.41
Sharpe Ratio	17.38	7.21	16.93	−0.49	15.18	−0.76
Sortino Ratio	86.57	13.36	47.43	−0.69	32.43	−1.31
Information Ratio	9.56	5.43	9.40	−0.41	9.16	−0.63
Win Rate	0.87	0.66	0.88	0.52	0.85	0.43
Profit Factor	28.95	3.20	24.61	0.93	14.86	0.89
Trade Expectancy %	0.49	0.27	0.48	−0.02	0.45	−0.03

the 1-step horizon, the optical RC achieves dramatically lower MAE (31.56 vs. 165.26), RMSE (38.47 vs. 207.23), NRMSE (0.10 vs. 0.56), and MAPE (0.20% vs. 1.06%) than ML, along with an R^2 of 0.99, indicating near-perfect fit. This performance persists even at longer horizons, where optical RC continues to outperform with lower forecast error, higher R^2, and substantially better directional accuracy (93.94% vs. 59.60% at 4-step and 79.80% vs. 54.55% at 10-step). The RC model consistently provides higher cumulative and annualized returns with lower drawdowns and stronger risk-adjusted performance, particularly as evidenced by Sharpe, Sortino, and Calmar ratios. From a trading perspective, the optical RC model proves significantly more profitable and consistent. At the 10-step horizon, it achieves a Profit Factor of 14.86 and a Trade Expectancy of 0.45%, compared to just 0.89 and –0.03% for the ML model, respectively. The Win Rate remains above 85% across all horizons, while ML falls below 50% at the 10-step mark. Additionally, the RC model delivers high Information Ratios throughout, indicating strong alpha generation relative to residual risk. Under the condition of limited input data, the ML model fails to generalize beyond the 1-step horizon, with diminishing returns, negative Sharpe and Sortino ratios, and poor directional accuracy as forecast length increases. In general, the optical RC model demonstrates superiority across multiple statistical and trading performance metrics. These results highlight the optical RC model's robustness and scalability, making it a compelling choice for financial forecasting and algorithmic trading on broad market indices like ^NYA.

To evaluate the predictive performance of the proposed optical RC model relative to the best baseline ML regressor across multiple financial indices and forecasting horizons, we computed the Mean Absolute Scaled Error (MASE) and its relative reduction. We adopt the MASE to enable scale-invariant and interpretable comparisons across assets and forecast horizons. By normalizing against a naïve forecast, MASE offers a robust, unit-free evaluation that accounts for series-level variability, making it particularly well-suited for financial time series characterized by nonstationary dynamics. For a predicted sequence $\hat{y}_t$ and true values y_t, the MASE is computed as:

$$\text{MASE} = \frac{\frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|}{\frac{1}{n-1}\sum_{t=2}^{n}|y_t - y_{t-1}|}$$

This formulation standardizes the forecast error by the average one-step naive forecast error (computed from the in-sample training data).

For each index and forecasting step, we compared the optical RC model's MASE against the best-performing baseline ML model. The relative MASE improvement is defined as:

$$\Delta_{\text{MASE}} = \frac{\text{MASE}_{\text{ML}} - \text{MASE}_{\text{RC}}}{\text{MASE}_{\text{ML}}}$$

To assess statistical significance, we employed bootstrap resampling with 1,000 iterations. In each iteration, residuals were sampled with replacement to estimate the distribution of relative improvement. We then computed the 95% confidence interval (CI) and a two-tailed p-value to determine whether the observed improvement was statistically significant. Significance levels were annotated using conventional symbols: $*$ for $p < 0.05$, $**$ for $p < 0.01$, and $***$ for $p < 0.001$.

The results were visualized using a heatmap in Fig. 8, where rows represent forecasting steps and columns represent financial indices. Each cell displays the mean relative improvement in MASE from the optical RC model over the best-performing ML baseline regressor, along with bootstrap-derived statistical significance markers. This bootstrapped simulation, using synthetic financial data with controlled noise, serves to rigorously validate the statistical robustness of our method prior to real-market deployment. For each index-step pair, training and testing data were synthetically generated to mimic plausible return dynamics. Gaussian noise was added to generate predictions from both the optical RC model and baseline regressors, simulating controlled levels of error.

Figure 8 illustrates that across the majority of indices and forecasting horizons, the optical RC model consistently outperforms the baseline ML regressors. Notable gains are observed on the ^N225 index with a 29.40% MASE

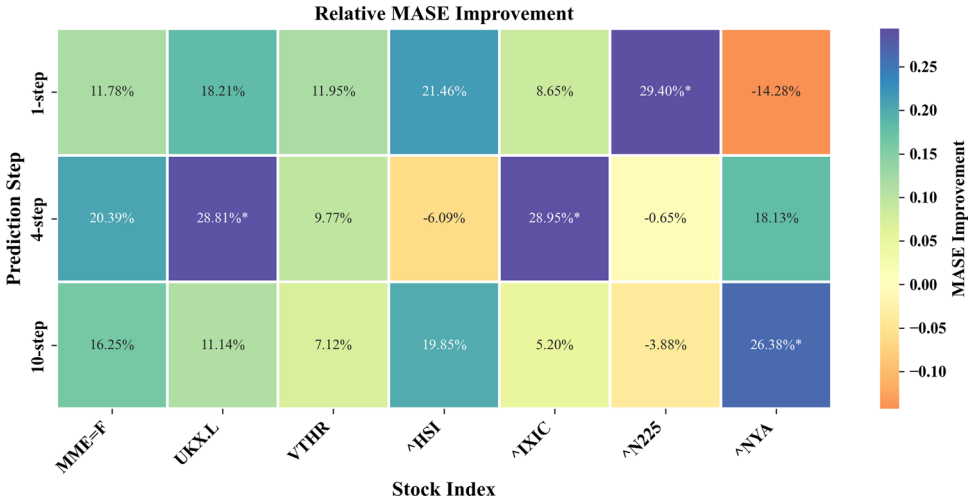


Fig. 8 Heatmap showing the relative MASE improvement of the optical RC model over the ML baseline regressor

reduction at the 1-step horizon ($p < 0.05$) and on the ^IXIC index with a 28.95% improvement at the 4-step horizon ($p < 0.05$). The strongest gains tend to occur at the 4-step horizon, suggesting that the optical RC model is particularly effective at capturing intermediate-range temporal dependencies. This aligns with the expected advantage of optical RC systems in modeling nonlinear dynamics and memory effects over medium-term windows. Although a few negative values are reported-such as ^NYA at the 1-step horizon (−14.28%) and ^HSI at the 4-step horizon (−6.09%)-these instances are isolated and not statistically significant, underscoring the robustness of the optical RC model across most experimental conditions.

These findings collectively support the effectiveness of optical RC as a scalable and statistically reliable alternative to conventional ML models in noisy, data-limited financial forecasting tasks.

Besides, we propose another way to quantify the performance difference by calculating the error reduction (ER) between the best ML model and optical RC predictions for seven stock indices. The ER is calculated using the formula

$$\mathrm{ER} = 1 - \frac{\mathrm{NRMSE}_{RC}}{\mathrm{NRMSE}_{ML}}, \tag{3}$$

where NRMSE_{RC} and NRMSE_{ML} are the NRMSE of the RC and the best-performing ML methods, respectively.

Figure 9 presents the relative error reduction achieved by the optical RC model in comparison to the best-performing ML model trained for each stock index. The results are shown for three forecast horizons: 1-step, 4-step, and 10-step predictions. The values represent the percentage decrease in Normalized RMSE, with higher values indicating greater relative improvement by the optical RC model. Notably, for ^NYA, optical RC outperforms the best ML models by over 80% at the 10-step and 4-step horizon and maintains relatively high margins at 1-step levels. Comparable results are observed for VTHR, UKX.L, and MME=F, all of which show reductions exceeding 50% across nearly all horizons. For some indices like ^IXIC and ^N225, the improvements are more modest, particularly at longer horizons. For example, the 10-step error reduction on ^N225 is just 27.82%, 1-step error reduction is 17.13%, and dips to 13.62% at the 4-step level, which is due to the higher market volatility. It suggests that even the best-trained ML models perform competitively in these particular markets. Nonetheless, optical RC still offers meaningful gains even in these more challenging contexts. Interestingly, the trend across most indices shows increasing relative gains with prediction horizon. This is significant given that multi-step forecasting typically suffers from error accumulation - yet optical RC maintains or even strengthens its advantage, demonstrating resilience and temporal consistency. This makes it a particularly attractive model for longer-term financial forecasting, where robustness and stability are critical.

To gain a deeper understanding of the accuracy spread and stability of our model, we plot the Absolute Percentage Error (APE) distributions in Fig. 10 for the optical RC and ML models across 1-step, 4-step, and 10-step forecasting

 https://doi.org/10.1007/s44196-025-00906-4

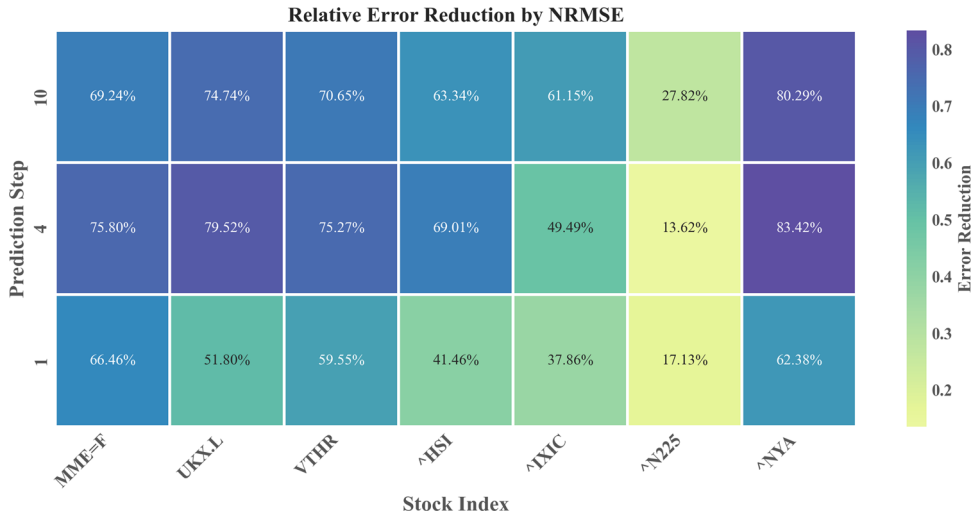


Fig. 9 The error reduction of the weighted optical RC compared to the best ML model across multiple stock indices

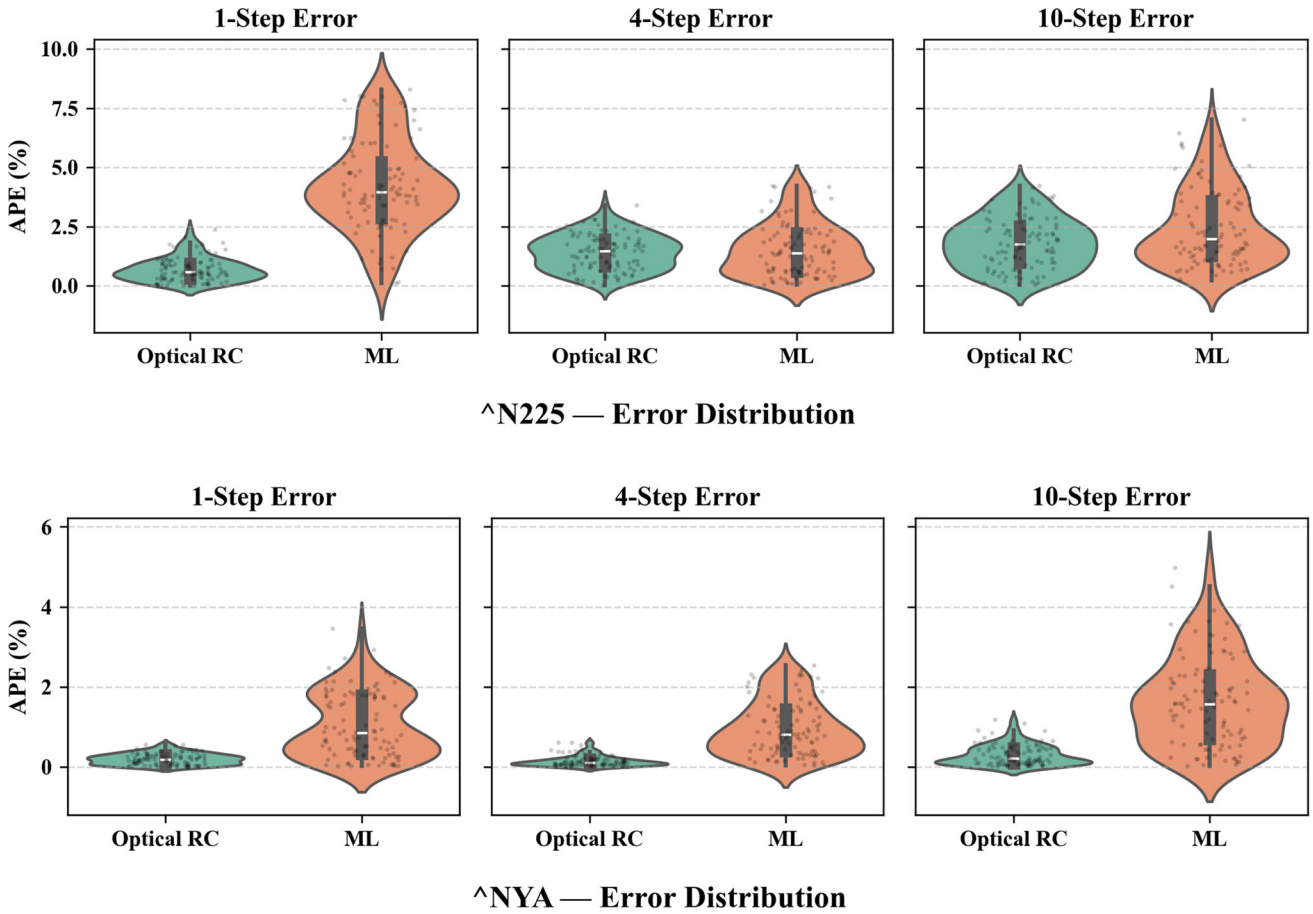


Fig. 10 Multi-step APE (%) distributions for stock indices ^NYA and ^N225

horizons. The APE is calculated by

$$\text{APE } \% = \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100\%$$

where y_t represents the actual value at time t, while $\hat{y}_t$ denotes the predicted value. The absolute percentage error (APE%) quantifies the relative deviation between the predicted and actual values as a percentage.

For the ^N225 index, the optical RC model consistently exhibits lower median errors and tighter distributions. At the 1-step horizon, optical RC errors are sharply concentrated around 0%, while the ML model demonstrates a wider, right-skewed distribution with a notably higher median. This trend persists as the forecast horizon extends: the optical RC model retains a compact error profile even at the 10-step horizon, while the ML model's errors become increasingly dispersed and skewed, indicating greater uncertainty and instability in its predictions.

A similar but more pronounced pattern emerges for the ^NYA index. The optical RC model produces remarkably low and stable error distributions across all horizons, with a median APE consistently near zero. In contrast, the ML model shows broader and more variable error distributions, particularly at the 10-step horizon where error magnitudes and outliers increase significantly. These results highlight the optical RC model's robustness in multi-step forecasting, offering not only improved accuracy but also significantly reduced variance, which is critical in financial contexts where reliability and stability are paramount.

6 Conclusion

In conclusion, we have established a weighted optical reservoir computing system in free space that effectively captures broader market behavior and demonstrates superior performance in stock index prediction. Compared to the existing machine learning methods and long short-term memory neural networks, our system offloads intense computation to an optical system with low energy consumption, showing significant advantages in handling high volatility and nonlinear market behaviors using limited data. This method promises a cost-effective approach to real-time, parallel, multi-dimensional data processing and predictions, with broad applications in financial forecasting. The next steps include moving more data processing into the optical domain for high-speed, high-quality, and versatile applications in financial analysis and alpha research. Future work could explore extending RC architectures through hierarchical or multi-scale reservoirs to better capture long-term dependencies and multi-scale temporal patterns in financial time series analysis.

Author Contributions Fang Wang, Ting Bu, and Yuping Huang contributed to the conceptualization of the study. Fang Wang and Ting Bu developed the methodology, conducted the formal analysis, and carried out the investigation. Validation was performed by Fang Wang, Ting Bu, and Yuping Huang. The original draft was prepared by Fang Wang and Ting Bu, while all three authors participated in the review and editing process. All authors have read and approved the final version of the manuscript.

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Data Availability The data underlying the findings presented in this study are publicly available and can be accessed via the GitHub repository at: (https://github.com/TingBu-forever/Supplymentary).

Declarations

Conflict of interest The authors declare that there is no conflict of interest.

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